TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*IMP110 – Impedance controlled PCB's up to 110GHz*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (..), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), **Next-gen Communication (√),** Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical and Electronics		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	Electrical and Electronic Components	*PCB components*	R&D / Engineering Services	Production Process Modeling / Simulation
Concept of Proposed Technology	*A key development of this research work is to establish new fabrication and verification procedures for PCB-manufacturer to provide a guaranteed service for frequencies up to 110GHz.* *PCB (Printed Circuit Boards) manufacturing is an industry service that deals with processes such as plating and photolithographic etching, drilling and layer lamination and pressing to produce multilayer PCB's based on organic substrates. Although organic substrates for high frequencies are available today with dielectric properties characterized up to 110GHz, PCB manufacturer hesitate to provide accurate and guaranteed services up to these frequencies. It would require deep understanding and enormous investments in high-frequency design tools and measurement equipment to implement such service. Instead of that, the risk is on the circuit designer. But designers have limited understanding and limited view into fabrication details. So, many high-frequency designs require several fabrication cycles, before they achieve sufficient performance.*			

Accurate high-frequency PCB design is predominantly dependent on the deep understanding of manufacturing processes and tolerances and materials by PCB designer, who don't have the processes in their hands. Todays workflow requires an inside view in manufacturing details, that are not shared with the designer.

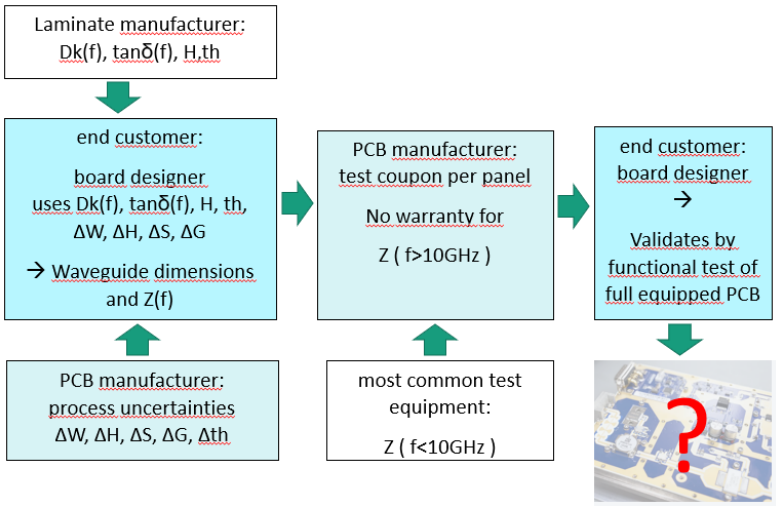

Figure 1: classic design and fabrication workflow

While PCB manufacturer provide services with controlled-impedance today up to max. 40GHz based on computation with waveguide-computation models based static field distribution and test their test-coupons with equipment from POLAR Instruments ode Anritsu

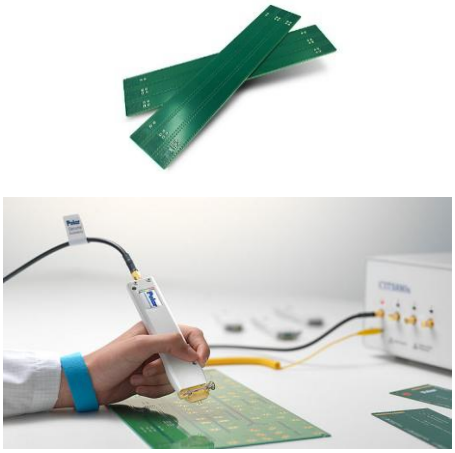

Figure 2: POLAR Instruments < 1GHz (most common and affordable)

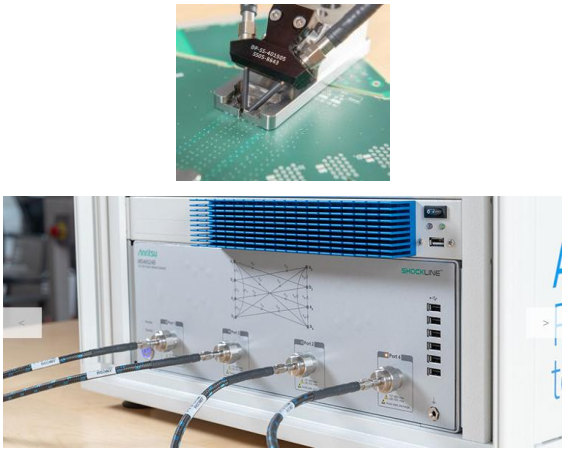

Figure 3: Anritsu 40GHz (expensive, rarely used)

They key development is to break this classic workflow and add knowledge from third party, the research institute, to provide pre-tested structures, choice of materials and waveguide types with pre-characterized performance to reduce design cycle and guarantee performance. Strong collaboration between PCB manufacturer and research institute is necessary to adopt specific process details in the simulation models.

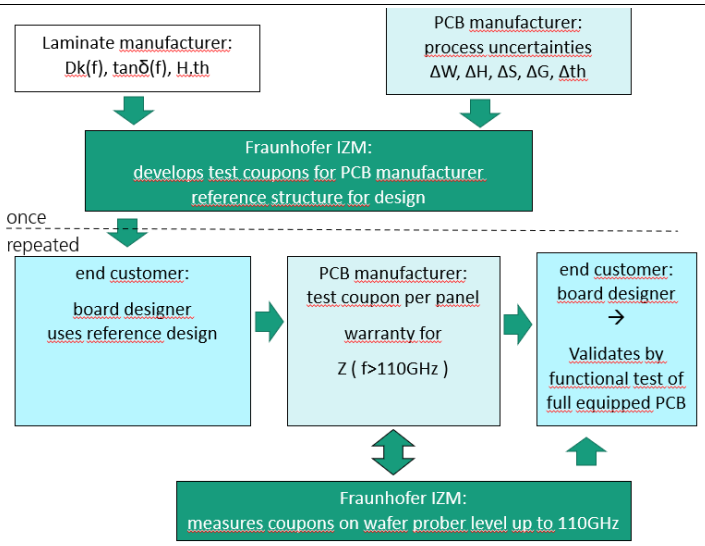

Figure 4: new workflow, Fraunhofer is supporting Korean PCB manufacturer that can provide guaranteed HF performance

Fraunhofer IZM has extensive knowledge about fabrication procedures from PCB to wafer-level processes. While PCB manufacturing is limited to structures (lines/spaces) of >50µm, Fraunhofer has shown metallization structures down to 3µm. Several research projects for 6G mobile communication in the range of 140GHz have shown the competence in packaging of IC's (Integrated Circuits) in projects such as 6G-KOM, 6G-RIC. Further research on 140GHz is ongoing in research projects such as 6G-TERAKOM and Xtreme-6G, providing a deep knowledge and experience with highest frequency design, dealing with modern PCB laminates such as Panasonic Megtron-8 and EMC (Epoxy Mold Compounds) materials regarding its high-frequency behavior and its processability.

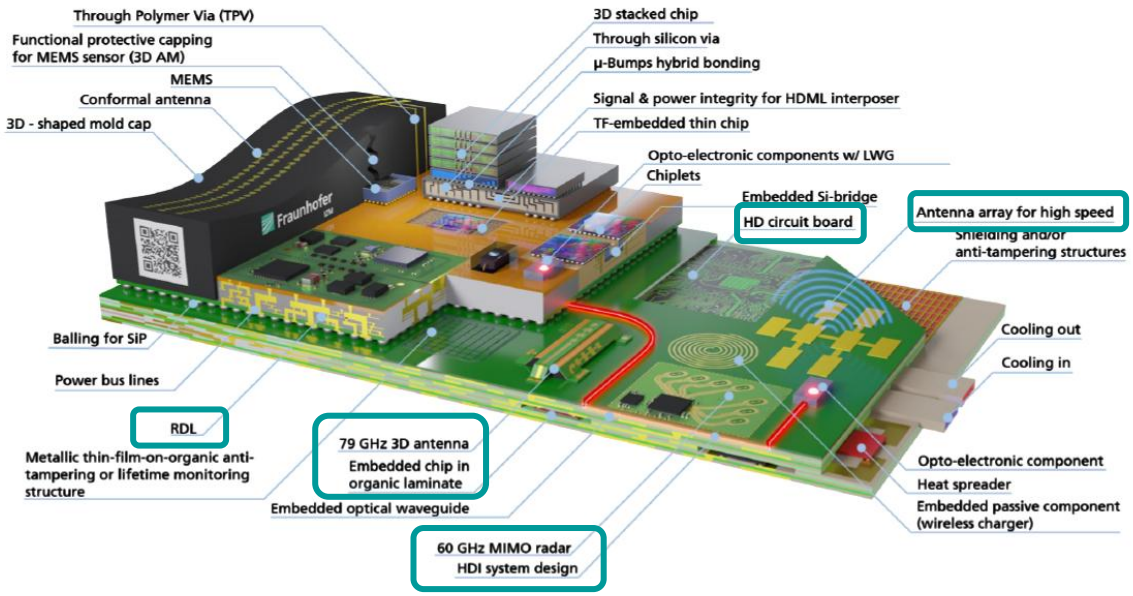

Figure 5: integration technologies at Fraunhofer IZM, relevant technologies marked

Figure 6: Substrate Integration Line at Fraunhofer IZM

With this knowledge test-structures were developed to control process parameters, such as over/underetching with fine-line structures, waveguides to control waveguide impedances, resonators and capacitors to monitor properties of the substrate that needs to be adopted and optimized to the processes of Korean PCB manufacturer. Fraunhofer uses 3D Maxwell simulation and improved analytical models to compute impedance matching against process tolerances.

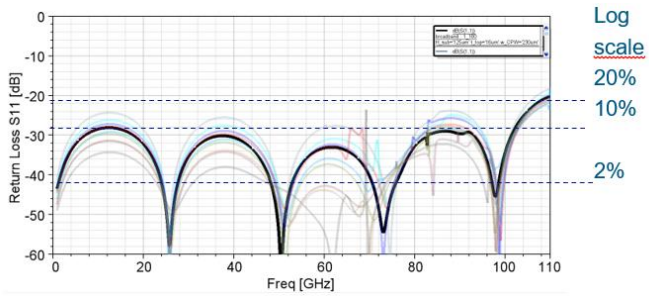

Figure 7: impedance matching (given as return loss) varying with process tolerances

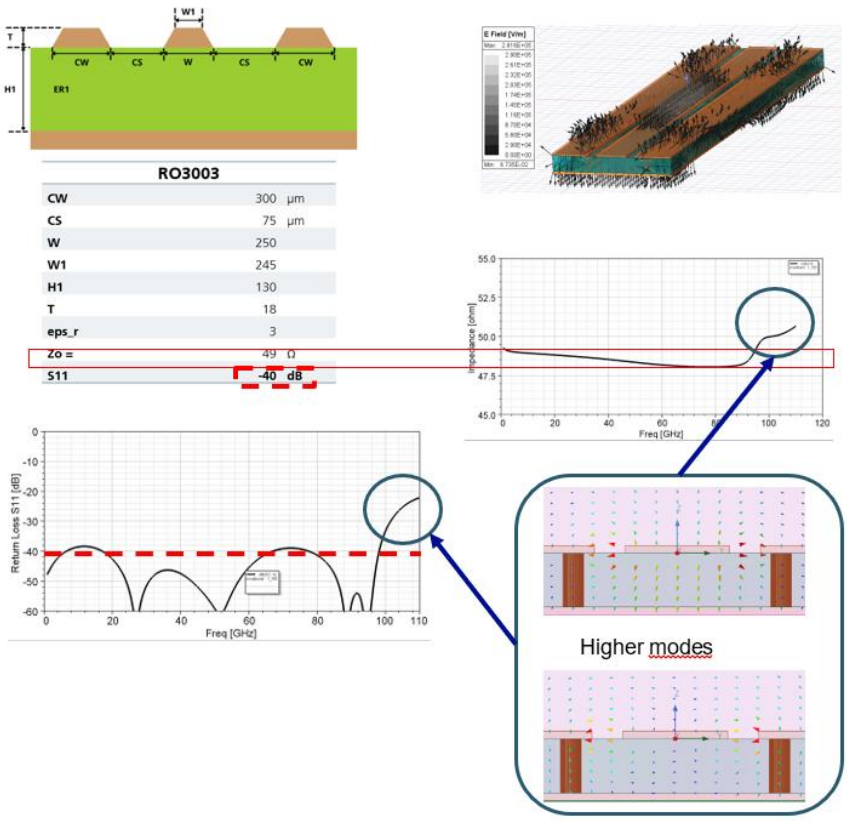

Figure 8: 3D Maxwell simulation provides information about frequency dependency and mode excitation that static models cannot do, these methodologies support Korean PCB manufacturer

Figure 9: Formfactor / Keysight automatic probing station (110GHz) at Fraunhofer IZM for volume testing

Rationale for Joint R&D Project	*Many of the new technologies, such as 5G mobile communication (28GHz today), future 6G mobile communication (140GHz), modern computer interfaces PCIe Gen.7 (128GT/s), opto-electrical interfaces Ethernet-CEI (224Gbd), automotive radar (79GHz) are emerging and require reliable PCB services today. Even at higher frequencies and data rates the impact of process tolerances is even more critical, e.g. operation frequency of antennas might drift due to over/under etching, electrical bandwidth reduces due to impedance mismatch.*

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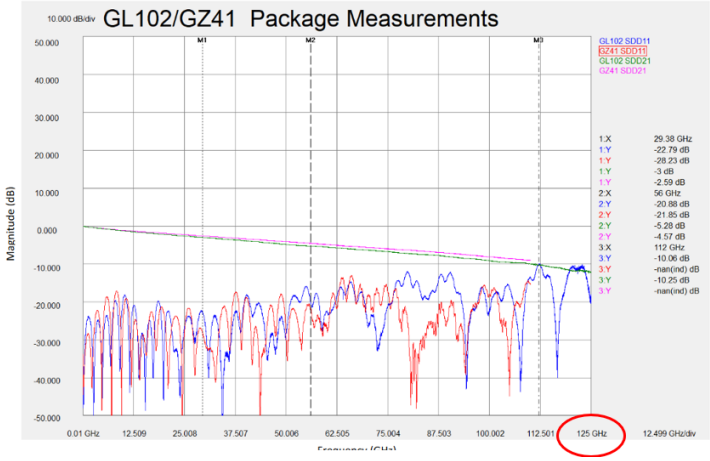

S11 [dB] specified up to 125GHz

Figure 10: required bandwidth for next generation 224Gb/s (OIF-FD-CEI-224G)

Today no PCB manufacture can guarantee HF performance of its PCB processes. The risk is on the customer, which is contrary to what customers are used in the past and what is required for the industry. Especially when it comes to volume fabrication, quality control is mandatory.
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1. high frequency applications, Ethernet 224Gbps, 5G / 6G mobile
2. ultra low loss, low Dk substrate materials
3. process tolerances in PCB manufacturing
4. high frequency dependent modelling and simulation
5. test coupon measurement up to 110GHz
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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (9)

R&D Objectives

Fraunhofer IZM will develop test coupons to monitor processes during manufacturing for the development phase of the process. This helps to identify and quantify process tolerances at Korean PCB manufacture that will impact the HF-performance. Fraunhofer IZM will further develop simulation models to predict the tolerance range and defines suitable application windows for combinations of waveguides and materials. At the end of the project well defined test coupons will be available the can be tested against specification, so that customers of Korean PCB manufacturer can verify the delivered quality and have a guarantee for the HF performance of their circuit.

The Korean PCB manufacturer will develop process monitoring schemes to provide data for process analysis that is required for proper modeling.

R&D Target Specifications

Suggested list of materials to be confirmed by the Korean PCB manufacturer

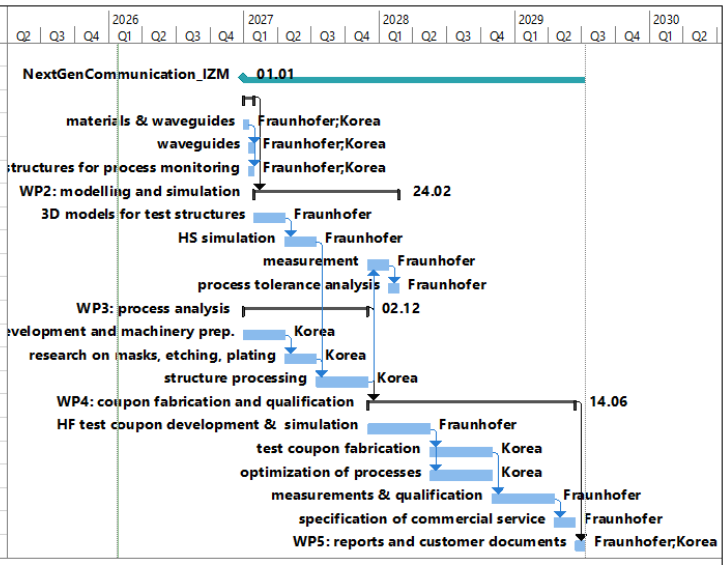
Laminate manufacturer	type	Specified max. frequency	Material type	Targeted application
		[GHz]		
Rogers	RO 3003	110	PTFE + ceramic	measurement grade
Panasonic	Megtron 8	110	Glass	industrial grade
Ventec	Tec-speed	110	PTFE composite	industrial grade
EMC	EM 892	80	Glass	measurement grade
Panasonic	Megtron 6	80	Glass	prototype level

Targeted tolerances depending on application class to be confirmed by the Korean PCB manufacturer

Targeted impedance	tolerance +/-	Resulting impedance	reflection |S11|	application
[Ω]	[Ω]	[Ω]	[dB]	
50	0,1	50,1	-30	measurement grade
50	1,0	51,0	-20	industrial grade
50	5,0	55,0	-13	prototype level
100	0,1	100,1	-33	measurement grade
100	1,0	101,0	-23	industrial grade
100	5,0	105,0	-16	prototype level

R&D Plan

Nr.	Vorgangsname
1	
2	**NextGenCommunication_IZM**
3	**WP1: Specification**
4	materials & waveguides
5	waveguides
6	test structures for process monitoring
7	**WP2: modelling and simulation**
8	3D models for test structures
9	HS simulation
10	measurement
11	process tolerance analysis
12	**WP3: process analysis**
13	process development and machinery prep.
14	research on masks, etching, plating
15	structure processing
16	**WP4: coupon fabrication and qualification**
17	HF test coupon development & simulation
18	test coupon fabrication
19	optimization of processes
20	measurements & qualification
21	specification of commercial service
22	WP5: reports and customer documents



WP1 specifies materials, waveguide types, targeted application levels, lines/spaces, tolerance parameters such as thicknesses of dielectric and copper layer, finishes such as Gold/Silver.
WP2 Fraunhofer develops HF test structures for process monitoring and refines existing models for proper tolerance predictions. After fabrication of the structures, Fraunhofer will measure the structures on several panels to acquire statistical data and feed models for deeper investigation.
WP3 Korean SME will investigate process tolerances and develops monitoring schemes and methods. The gathered data will be used for modelling at Fraunhofer. User feedback will be evaluated to define target performance specification.
WP4 Fraunhofer develops final HF test coupons used for fabrication verification. Korean SME will produce panels with such structures that will measured at Fraunhofer. Concluding from these results a specification will be developed.
WP5: assess technical achievements, generate customer documents such as data sheets
MS1: models developed, processes analyzed, end of WP3
MS2: final test coupons analyzed and achievements specified, end of WP4

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Fraunhofer IZM* *develops test structure for process monitoring, develops models to predicts tolerance spread* *develops test coupon for customer verification* *Measures all structures, derives achievable max. frequencies and application class*
Expected R&R of Korean SME Partner	*A transfer of knowledge from Fraunhofer IZM to Korean SME how to achieve and guarantee high-frequency performance of PCB's during production. Korean PCB manufacturer processes the test structures, generates statistical data for its production, develops monitoring procedures that generate data for modeling and simulation.* *Potential partners are Interflex, Mtech, Terranix, JS-PCB or SEP.* *The Korean SME will validate the developed processes and refine them to a commercialized level.*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Fraunhofer IZM will generate knowledge about PCB processes at Korean SME, develop models for simulation and develops test structure for process monitoring and product qualification.* *The Korean SME will be able to provide PCB's with guaranteed performance up to 110GHz. Test coupons will provide proof of quality for the manufacturer and its customers. This strengthens the competitiveness on international level, since such services are not yet available at that scale.*

6. References

(1) ECTC, 2025, T. Lim, S. Kosmider, K. S. Murugesan „Integration of D-Band 140 GHz III-V Components Transceiver in Embedding PCB-Based Technology"

(3) IEEE, 2024, Prof. Ndip, Dr. Koeszegi, Jens Schneider „Temperature dependent Dielectric Characterization of Low Loss Thin Film Polymer up to sub-THz bands"

(3) EPTC, 2023, Dr. Tschoban, L. Becker "Analysis of the influences of PCB process tolerances and assembly tolerances on 60 GHz radar sensor for Radar toolkit"

(4) ECTC, 2022, K Zoschke, "High Density Thin Film Flex Technology for Advanced Packaging Applications"

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization